
Channel-Set Drift in Long-Reasoning W4A16: Why Static Protection Fails and When Rotation Helps

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Abstract

Static per-channel protection is attractive for W4A16 reasoning inference, but it assumes that high-magnitude activation channels remain useful protection targets throughout decode. We measure decode-time channel-set drift across four reasoning LLMs and find strict top-1% set-leaving of 53–67% by horizons up to 20K tokens, with a MATH-500 replication within 0.02 of the AIME-2025 rates on three measured models. We observe the effect at the measured block-output surface in hybrid Mamba-2, parallel-hybrid, and pure-Transformer models, extending DecDEC’s short-horizon Transformer observation and showing that Quamba2’s channel-order and activation-persistence observations do not imply stable top- K block-output protection at this measured surface. We then test eight channel-set protection families, a ParoQuant-style rotation baseline, rotation-plus-budget composition, and a small set of drift-aware rotation follow-ups. The results are regime-dependent. Hard position or scale switches are actively harmful, top-1% smoothing and cross-tensor coupling have signal but insufficient recovery, and reactive DecDEC-style selection does not recover the W4A16 gap. Budgeted EMA succeeds only in the Nemotron MoE-hybrid regime: top-10 EMA recovers 0.815 of the BF16-vs-static gap with a 0.220 descriptive margin over static top-10. Rotation is the strongest current baseline: ParoQuant-style recovery is 0.754 on Granite-Small, 1.05 on Nemotron-3, 0.756 on DeepSeek (high median, negative lower CI), and 0.381 on Falcon-H1. Drift-aware tight clipping improves Granite’s recoverable rotation tail (median 0.754 to 0.922; worst trace -26.37 to 0.601), but it does not transfer cleanly to DeepSeek or Falcon; residual correction and complete Granite internal-surface diagnostics also fail their promotion gates. We derive a prospective calibration protocol from the four-model study and evaluate its decisions descriptively; held-out validation of the thresholds remains future work.

1 Introduction

Long-reasoning models can spend thousands of decode steps in a single answer, so calibration-time quantization decisions must survive within-trace distribution shift. Static protected-channel maps are especially brittle if the channels that dominate early activation magnitude do not remain dominant later. This paper studies that assumption for W4A16 post-training quantization (PTQ): 4-bit weights, 16-bit activations, and a small protected activation-channel set kept at higher precision.

The assumption matters for recent state-space quantization. Quamba2 reports channel-order preservation in selective-SSM computation and channel/state persistence for SSM activations. Those observations motivate its offline sort-and-cluster weight reordering and per-state-group quantization (Chiang et al., 2025). We test the corresponding protected-set assumption at long reasoning horizons. Across Granite-4.0-H-Small, NVIDIA-Nemotron-3-Nano-30B-A3B-BF16 (Nemotron-3), DeepSeek-R1-Distill-Qwen-1.5B, and Falcon-H1, 53–67% of the top-1% channels at the calibration position strictly leave the later top-1% set. The channels do not merely shuffle rank within the set; they leave the set a static policy would protect.

We target a narrow but practical systems problem. Reasoning workloads are expensive, and inference-cost forecasts expect large pressure to reduce per-token cost even as reasoning demand grows (Wiggers, 2025; Gartner, 2026). Recent low-bit PTQ systems for larger standard Transformers can recover reasoning quality, including ParoQuant’s weight-only scaled-pairwise-rotation reasoning results (Liang et al., 2026). Our focus is different. We study small reasoning models, hybrid Mamba-2 or parallel-hybrid architectures, pure PTQ rather than distillation, and channel-set protection under long decode.

Our contribution is diagnostic and prescriptive: we identify a measured failure mode and derive a calibration protocol for choosing channel-set protection, budgeted smoothing, rotation, or rejection. The absence of a new universal quantizer is part of the finding: in these regimes, rotation absorbs much of the recoverable channel-set signal, while residual columns and internal-surface placement do not add a promoted remedy.

Our contributions are fivefold.

- Decode-time protected-channel drift. We measure protected-channel set drift across four models and architectures up to 20K tokens, finding strict set-leaving of 53–67% across hybrid Mamba-2, parallel-hybrid, and pure-Transformer models, with MATH-500 replication on three models.
- Matched-control failure taxonomy. We evaluate static maps, hard switching, binned scales, EMA smoothing, cross-tensor coupling, reactive selection, larger-budget tracking, stable cores, predictive tracking, and residual/surface screens with random or matched-cost controls where applicable.
- Mechanism: identity drift vs. basis removal. The channel-set failures point to unstable channel identity, harmful discontinuities, insufficient top-1% capacity, and model-dependent budget headroom. ParoQuant-style rotation is the strongest baseline because it removes much of the basis dependence instead of chasing original-basis channel identities.
- Regime map and calibration guidance. Budget-tuned EMA succeeds on Nemotron at top-10, recovers 0.449 on Granite top-5 with wide uncertainty, and remains ambiguous on DeepSeek and Falcon. ParoQuant-style rotation recovers 0.754 on Granite, 1.05 on Nemotron, 0.756 on DeepSeek, and 0.381 on Falcon. We turn these outcomes into a descriptive calibration checklist that provisionally suggests rotation, budgeted EMA, tail-control, or rejection of channel-set protection by measured regime.
- Residual-correction systems bound. A valid top-8x32 rotated-basis residual proxy worsens the Granite tail gate, so residual energy is not a reliable selector. We still quantify the HBM, MAC, working-set, and energy envelope that any future residual-correction method would need to justify.

2 Related Work

Mamba and state-space quantization. Vision-Mamba quantization already reports dynamic activation outliers. QMamba introduces temporal grouping for hidden states (Li et al., 2025), and OuroMamba uses dynamic outlier detection during inference (Ramachandran et al., 2025). Language-Mamba work often assumes more stable structure. Mamba-PTQ identifies recurrent outlier channels (Pierro and Abreu, 2024); Quamba and Quamba2 use static or calibration-time structure for Mamba-family PTQ (Chiang et al., 2024; 2025); MambaQuant uses variance-aligned rotations (Xu et al., 2025). We ask whether the static protected-set assumption holds in long reasoning traces.¹

LLM quantization and rotations. SmoothQuant, AWQ, QuaRot, KVQuant, and BlockDialect protect or transform activation structure using calibration-time statistics, rotations,

¹We use decode-time channel-set drift for the measured phenomenon. SmoothQuant’s “migration” is activation-to-weight scale transfer (Xiao et al., 2023); MoBiQuant uses related language for token-set shifts across bit-widths (Wang et al., 2026).

or adaptive formats (Xiao et al., 2023; Lin et al., 2023; Ashkboos et al., 2024; Hooper et al., 2024; Jang and Tambe, 2025). CMPQ allocates per-channel precision from calibration-time activation distributions; our work tests whether calibration-time channel selection remains stable under long-decode drift (Chen et al., 2025). ResQ keeps a low-rank PCA subspace at higher precision throughout decode, while we measure whether channel-basis top-K protection is stable across decode positions and find it often is not (Saxena et al., 2025). Rotated Runtime Smooth is the closest rotation-plus-runtime-smoothing adjacency. It uses runtime channel/group-wise activation maxima as smoothing scales in fused GEMM rather than maintaining per-layer top-K EMA protection sets (Yi et al., 2024). Our results do not contradict these systems. The narrower implication is that, when protected channel sets change during long decode, static channel maps need architecture- and horizon-specific validation. ParoQuant is particularly important because it reports strong reasoning-task accuracy under 4-bit weight-only PTQ using scaled independent Givens/pairwise rotations plus channel-wise scaling (Liang et al., 2026); its project page reports Qwen3-4B AIME-24 scores of AWQ 62.2, ParoQuant 73.3, and FP16 75.6 (Z Lab, 2026). Our ParoQuant-style packets use rotation as a strong baseline, not as our method.

Prior rotation methods construct static calibration-time transforms, learned rotations, or runtime smoothing rules that improve low-bit inference by changing the basis in which quantization error appears. Our question is different: under long reasoning decode, channel identity itself drifts. We measure when channel-set protection becomes ill-posed, then test whether rotation removes that basis dependence or whether drift-aware tail control, surface choice, branch-local rotation, or rotated-basis residual correction can improve on a fixed ParoQuant-style baseline.

Dynamic and long-context policies. DecDEC dynamically identifies salient activation channels at each decode step and reports static-channel recall around 30% or below against per-step ground truth in short Transformer decode (Park et al., 2025). We extend that setting in four ways: longer reasoning horizons, strict set-membership drift accounting, hybrid Mamba/parallel-hybrid coverage, and reasoning-trace intervention controls. The pure-Transformer result serves as a falsification/control surface. KV-cache methods such as KIVI, PM-KVQ, AttentionPredictor, ChanMix, and MixKVQ adapt or compress cached key/value tensors (Liu et al., 2024; 2025; Yang et al., 2025; Liao and Wen, 2026; Zhang et al., 2025); our interventions target linear-layer activation channels.²

Error accumulation. SLQ shows that layer-wise non-uniform/asymmetric quantization with wide bitwidth search can be task-lossless below 4 bits per parameter and distribution-lossless at roughly 5–6 bits per parameter (Helwig et al., 2026). QEP emphasizes cross-layer quantization-error growth (Arai and Ichikawa, 2025). LAQuant emphasizes layer-wise look-ahead losses and calibration/deployment alignment for reasoning models (Choi et al., 2026). Activation Sensitivity formalizes channel-wise perturbation impact on loss and downstream propagation (Xu, 2026). These results motivate testing whether our more aggressive W4A16 endpoint exhibits decode-position error growth rather than assuming it.

3 Method

For each trace, layer, and measured decode position, we record the vector of per-channel activation magnitudes. Let $a_{\ell,p,c}(t)$ be the activation magnitude for layer ℓ , prompt p , channel c , and decode position t . The population channel score is $\sigma_{\ell,c}(t) = \mathbb{E}_p[|a_{\ell,p,c}(t)|]$, and the ideal top-K protected set at position t is

$$S_{\ell}(t) = \arg \text{topK}_{c \in \{1, \dots, d_{\ell}\}} \sigma_{\ell,c}(t).$$

²“Hot channels” in HCP/CHON refers to persistently extreme channels during NVFP4 pre-training (Dong et al., 2026). This training-step time axis complements our decode-step time axis: channels can become persistent across pretraining iterations while drifting during a single inference trajectory. We refer to Koike-Akino et al. (2026) as test-time TTQ to avoid conflict with classical Trained Ternary Quantization.

Static channel protection assumes $S_\ell(t) \approx S_\ell(t_0)$ for later decode positions. Our empirical estimator keeps prompt identity explicit: let $S_{\ell,p}(t)$ be the top-1% channel set for layer ℓ and prompt p at position t . The strict set-leaving rate at position t is

$$1 - \frac{|S_{\ell,p}(100) \cap S_{\ell,p}(t)|}{|S_{\ell,p}(100)|}, \quad (1)$$

averaged over prompts and layers. Within-set shuffling is reported separately for channels that remain in the top-1% set but move substantially in rank.

We score interventions using recovery of the BF16-vs-static W4A16 gap:

$$R = \frac{L_{\text{static}} - L_{\text{method}}}{L_{\text{static}} - L_{\text{BF16}}}. \quad (2)$$

Recovery may be negative when a method is worse than static protection. It may exceed one when a method beats BF16 on the scored slice. We report unclipped medians and bootstrap CIs. We explicitly track traces with no recoverable static gap and exclude them from ratio computation for method CIs; their frequency is part of the trace-heterogeneity result.

The method classes are grouped by the mechanism they test: static coverage, hard switching, smoothing, cross-tensor signal, reactive selection, larger-budget tracking, stable cores, and rotation. We test eight channel-set method classes: static migration-aware unions, hard position-conditioned protected sets, hard position-binned scale tables, EMA-smoothed top-magnitude protection (M11), cross-tensor activation+key coupling (M18), DecDEC-style reactive top-1% selection, budget-tuned EMA protection (M11b), and stable-core protection (M26). We compare each method to static baselines plus random or matched-budget controls when applicable. We evaluate ParoQuant as an algorithmic rotation baseline, not as a channel-set method, and test one direct ParoQuant+M11b composition after both individual mechanisms showed signal.

The later DriftRot follow-ups are diagnostic screens rather than headline method claims. Tight clipping changes only the ParoQuant-style scale clamp and is promoted only if it improves held-out tail risk without losing the median. Rotated-basis residual correction tests a protected-column correction $y = xW_{\text{pq}} + x_P(W_{\text{fp}} - W_{\text{pq}})_P$ after ParoQuant. M-SURFACE tests whether internal projection-input surfaces have lower strict set-leaving than the block-output surface. Each screen is compared against the fixed ParoQuant-style baseline, and failures are recorded as design constraints rather than hidden tuning attempts. Our ParoQuant packets are local, ParoQuant-style baseline implementations using the reported scaled pairwise-rotation design rather than official code; we use them to test whether rotation-class conditioning dominates channel identity tracking in the same local scoring harness.

4 Experiments

Models and traces. The measurement suite covers Granite-4.0-H-Small, Nemotron-3, DeepSeek-R1-Distill-Qwen-1.5B, and Falcon-H1. Intervention experiments use 12 deterministic long-reasoning traces except where stated otherwise; the original drift measurement uses 24 Granite-Small traces. We measure decode positions on the grid $\{100, 500, 1000, 5000, 10000, 20000\}$ when the model packet supports 20K tokens. The cross-prompt replication uses 30 MATH-500 prompts on Granite-Small, DeepSeek-R1-Distill, and Falcon-H1; it supports drift generalization, not recovery generalization. GPQA-Diamond was gated and not used for claims. Dense KL analysis uses a Granite-Small packet with all positions 1–512, every 10th position to 2K, every 50th position to 10K, and every 100th position to 20K. The narrowed cross-model diagnostic repeats the KL/FFT summary on DeepSeek-R1-Distill and Falcon-H1.

Controls. Static top-1% protection is the baseline. Position- and scale-switching methods use random-bin controls. EMA and reactive methods use random-walk or random-reactive controls. Budget experiments include matched static top-10 controls. The controls are decisive: two methods that look plausible in isolation are worse than their random controls.

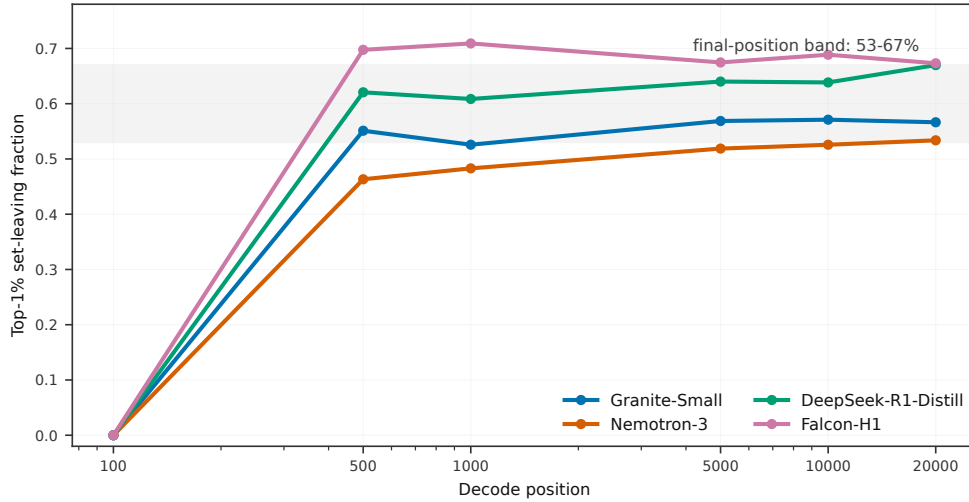


Figure 1: Strict top-1% channel-set leaving relative to decode position 100. Static channel protection loses 53–67% of its original protected set by the final measured position across four reasoning models.

5 Results

5.1 Channel sets drift at long reasoning horizons

Figure 1 shows the headline measurement. The traces are not monotone at every intermediate point. Even so, all four models have large strict set-leaving by the final measured position: 0.566 for Granite-Small, 0.534 for Nemotron-3, 0.671 for DeepSeek-R1-Distill, and 0.674 for Falcon-H1. The pure-Transformer reference drifts at least as much as the hybrid models, so the drift extends beyond Mamba layers.

The drift effect is not an AIME-only artifact in the models where we reran the measurement. On 30 MATH-500 prompts, strict set-leaving is 0.561 for Granite-Small, 0.651 for DeepSeek-R1-Distill, and 0.675 for Falcon-H1. Each is within 0.02 absolute of its AIME-2025 rate, passing the fixed cross-prompt tolerance. This replication is for channel-set drift; intervention recovery remains measured on the AIME trace packets. GPQA-Diamond was not included because the local benchmark access path was gated during this run.

5.2 Layer type does not explain the drift away

Because the Quamba2 comparison turns on persistence in SSM computation, the relevant observable here is whether drift reaches SSM/Mamba block outputs rather than only attention or MLP outputs. Figure 2 shows that Granite and Nemotron block-output drift is comparable across layer types. Granite attention and SSM/Mamba outputs have strict set-leaving of 0.557 and 0.567. Nemotron attention, MoE, and SSM/Mamba outputs have 0.563, 0.527, and 0.533. This does not isolate internal SSM state tensors, but it shows that the measured block-output surface does not preserve channels.

5.3 Top-1% channel-set interventions fail despite control signal

Figure 3 and Table 1 summarize the intervention outcomes. The random-bin control beats hard position-conditioned protection by 0.668 median recovery, and random scale-bin assignment beats hard scale bins by 0.761. EMA smoothing is different: M11 beats its random-walk control by 1.25 median, but recovers only 0.048 in absolute terms. M18 activation+key coupling has the strongest signal-vs-random margin among failed top-1% methods, beating random coupling by 4.32 median. Its absolute recovery is still negative. DecDEC-style reactive selection beats a random reactive control by 1.19 median, but still has negative

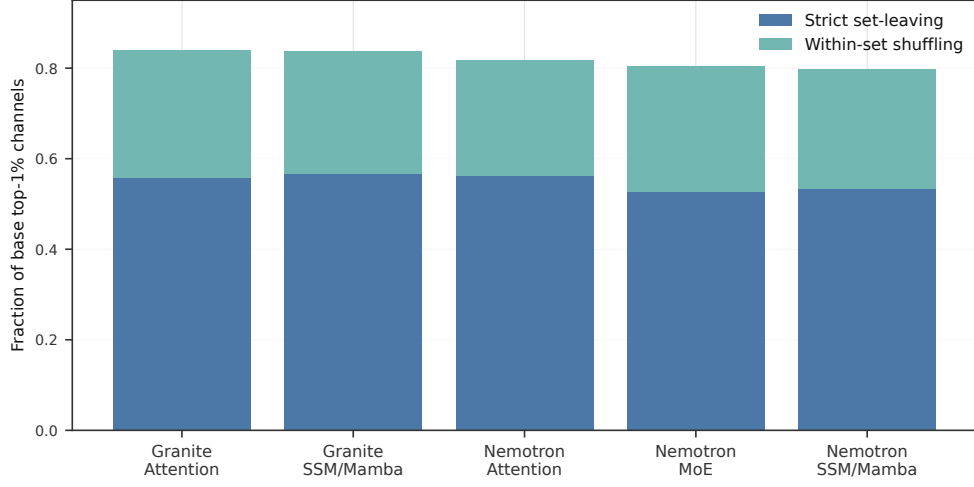


Figure 2: Layer-stratified drift at measured block outputs. Attention, SSM/Mamba, and MoE-classified layers show comparable strict set-leaving, so no single component type localizes the drift.

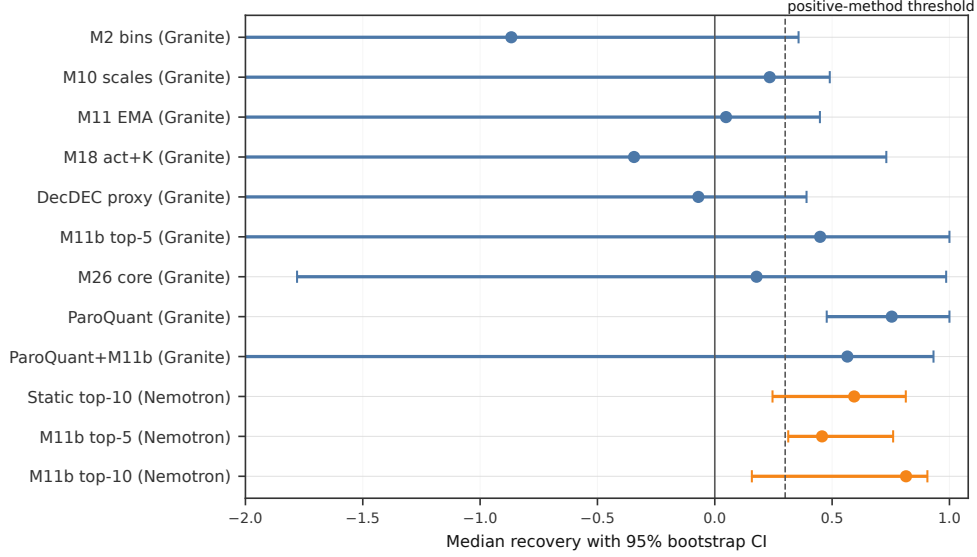


Figure 3: Median recovery with 95% bootstrap CIs. The plot clips long negative CI whiskers at the boundary for readability; Table 1 reports exact rounded values. The dashed line marks the 0.30 method-promotion threshold used in the method gates.

recovery and does not recover quality. M26 stable-core protection shows signal over random matched-core channels but remains below the method-promotion threshold.

5.4 Rotation is the strongest baseline; DriftRot gains are narrow

M11b turns a uniformly negative channel-set picture into a model-dependent positive result, while ParoQuant sets the stronger rotation baseline. On Granite-Small, top-5 EMA protection recovers 0.449 of the static gap and has a 0.501 descriptive median margin over static top-10, but the BCa CI is wide. Nemotron top-5 has a strictly positive BCa CI, yet it does not beat static top-10. The top-10 arm is the strongest channel-set result: median recovery 0.815, BCa CI [0.158, 0.906], Holm-adjusted $p = 0.030$, and a 0.220 median margin over

Method family	Model	Median recovery	95% CI	Key comparison
Position-conditioned channels	Granite	-0.867	[-4.25, 0.357]	random bin better by 0.668
Position-binned scales	Granite	0.234	[-3.47, 0.490]	random scale bin better by 0.761
EMA-smoothed top-1%	Granite	0.048	[-8.94, 0.448]	random-walk lower by 1.25
Activation+key coupling	Granite	-0.344	[-14.1, 0.731]	random coupled lower by 4.32
DecDEC proxy	Granite	-0.070	[-8.50, 0.391]	random reactive lower by 1.19
Stable core	Granite	0.178	[-1.78, 0.986]	random matched-core lower by 1.73
Budget EMA top-5	Granite	0.449	[-2.28, 1.00]	static top-10 lower by 0.501
Static top-10	Nemotron	0.594	[0.246, 0.814]	Holm-significant matched-budget control
Budget EMA top-5	Nemotron	0.457	[0.313, 0.760]	Holm-significant, below static top-10
Budget EMA top-10	Nemotron	0.815	[0.158, 0.906]	Holm-significant; static top-10 lower by 0.220
Budget EMA top-10	DeepSeek	0.335	[-0.409, 0.494]	ambiguous; static top-10 is 0.377
Budget EMA top-10	Falcon	0.044	[-0.144, 0.203]	ambiguous; static top-10 is -0.025
ParoQuant rotation	Granite	0.754	[0.477, 1.00]	M11b top-5 lower by 0.304
ParoQuant rotation	Nemotron	1.05	[1.01, 1.29]	median above M11b top-10 by 0.232
ParoQuant rotation	DeepSeek	0.756	[-0.246, 0.855]	median above static top-10 by 0.379
ParoQuant rotation	Falcon	0.381	[0.0645, 0.547]	median above M11b top-10 by 0.337
ParoQuant+M11b top-10	Granite	0.565	[-63.6, 0.932]	trails ParoQuant by 0.189
DriftRot tight clip	Granite pos-gap	0.922	[0.781, 1.65]	fixes ParoQuant tail; worst -26.37 to 0.601
DriftRot tight clip	DeepSeek	0.518	[0.265, 0.797]	median lower than ParoQuant by 0.232
DriftRot tight clip	Falcon	0.390	[0.0019, 0.637]	median flat; worst trace regresses

Table 1: Intervention outcomes with BCa 95% CIs where exact per-trace recovery values are available. We leave recovery unclipped; values can be negative or exceed one. The Holm tests evaluate median recovery greater than zero across the method-table family; paired method-vs-control margins are descriptive unless separately tested. ParoQuant is included as a rotation baseline and is not a channel-set method.

static top-10. Budget is a real mechanism, but not a universal remedy. The best budget and the value over static protection are model-dependent.

ParoQuant constrains the claim more strongly after baseline vetting. A ParoQuant-style rotation baseline recovers 0.754 on Granite-Small, 1.05 on Nemotron-3, 0.756 on DeepSeek-R1-Distill, and 0.381 on Falcon-H1. The Nemotron result is especially important because its median recovery is 0.232 higher than the previous M11b top-10 channel-set result on the same BF16/static baseline. Falcon also becomes a rotation-rescue case: the ParoQuant-style median is 0.337 higher than Falcon M11b top-10. DeepSeek has a high median but a negative lower CI, so we treat it as rotation-favorable but tail-risky. The paper does not argue that W4A16 PTQ is intrinsically broken; it argues that channel-set protection is fragile under long reasoning drift, while rotation-based conditioning often addresses the same endpoint by a different mechanism. That distinction matters because ParoQuant is a baseline, not our method.

The direct composition test was negative. On Granite-Small, ParoQuant+M11b top-10 recovers 0.565, trailing ParoQuant alone by 0.189 median recovery and also trailing the ParoQuant+random top-10 control. This rules out the simplest “stack the two remedies” story in the tested Granite composition packet. Rotation appears to absorb much of the recoverable channel-set signal in this setting, so future remedies need a decision rule or a more coupled design rather than naive composition.

The DriftRot follow-ups narrow the path further. Tightening the ParoQuant scale clip to [0.5, 2.0] fixes Granite’s recoverable positive-gap tail: across the eight recoverable Granite traces, median recovery rises from 0.754 to 0.922 and the worst trace moves from -26.37 to 0.601. The same setting does not transfer cleanly. On DeepSeek, the median drops from 0.756 to 0.518 and the worst trace worsens. On Falcon, the median is essentially flat, 0.381 to 0.390, while the lower CI and worst trace regresses. Rotated-basis residual correction also fails its current design gate: the valid top-8x32 MoE correction worsens the Granite tail trace from the tight-clip reference. This is more than a failed variant. It says residual energy is not aligned enough with loss benefit under the tested proxy; a residual method would need an output-impact selector rather than a norm-based column rule. A complete two-trace M-SURFACE diagnostic likewise fails to find a better placement surface. Granite post-block mean set-leaving is 0.546. The lowest internal means are SSM input after convolution (0.484) and SSM C generation after convolution (0.486), only about 0.06 below post-block and above the 0.30–0.40 promotion band; B generation, B/C concat, Mamba out-projection input, and attention out-projection input are 0.618, 0.602, 0.887, and 0.799. These failures are useful.

They say a stronger remedy cannot be simple clip retuning, residual columns selected by the current proxy, or internal-surface selection under the measured Granite packet.

5.5 A prospective calibration protocol

We derive a prospective calibration protocol from the four-model study and evaluate its decisions descriptively; held-out validation of the thresholds remains future work. On a new model, the protocol runs a short calibration packet and measures strict set-leaving, no-gap fraction, static top- K recovery, drift entropy/autocorrelation, and a rotation pilot if available. It then provisionally suggests static top- K , budgeted EMA, rotation, or rejection of channel-set protection for that regime; the suggested remedy still requires a local confirmation packet.

The descriptive decisions match the observed pattern. All four models exceed 0.53 strict set-leaving, so static top-1% protection needs validation rather than assumption. Nemotron passes the budgeted-EMA gate: M11b top-10 recovers 0.815 with CI [0.158, 0.906] and has median 0.220 above static top-10. The same gate is ambiguous on DeepSeek and Falcon: top-10 recovery is 0.335 with CI [-0.409, 0.494] on DeepSeek and 0.044 with CI [-0.144, 0.203] on Falcon. When a rotation baseline is available, the protocol compares it directly rather than stacking it with M11b. ParoQuant-style rotation is favored on all four current median comparisons: 0.754 on Granite, 1.05 on Nemotron, 0.756 on DeepSeek, and 0.381 on Falcon. DeepSeek remains tail-risky because its lower CI is negative. ParoQuant+M11b is sub-additive, so the protocol does not stack rotation and channel protection by default.

The same logic handles DriftRot screens. Tight clip is retained for Granite only after a tail-control check: it raises recoverable positive-gap median recovery from 0.754 to 0.922 and removes the catastrophic negative tail. The same setting is rejected as a universal retune because it lowers DeepSeek’s median and worsens Falcon’s worst trace. Residual correction and cheap projection-input M-SURFACE hooks do not promote until a new residual proxy or unmeasured internal SSM surface hypothesis is available.

This protocol is not a universal predictor. Its thresholds are derived from these four models, so the four-model evaluation is descriptive rather than held-out validation. The deployment checklist is still concrete: use rotation where it is locally measured; use budgeted EMA only where it clears a model-local matched-budget gate, as on Nemotron; apply tail-control retuning only when a calibration/confirmation split shows tail improvement, as on Granite; and reject channel-set protection when high drift is present but the matched-budget gate is ambiguous, as on DeepSeek and Falcon. The negative composition result is part of the guidance: remedies must be selected by measured model behavior, not stacked by default.

5.6 KL does not support runaway compounding

Figure 4 tests whether failed channel-set methods primarily fail because errors compound over decode. For per-position divergence $D_t = \text{KL}(P_{\text{BF16},t} \| P_{\text{Q},t})$, we fit three functional forms:

$$D_t \approx a + b\sqrt{t}, \quad D_t \approx a + bt, \quad D_t \approx a + bt^2.$$

The compound-error hypothesis predicts that linear or superlinear fits should dominate. It does not in this packet. In the dense Granite-Small packet, static top-1%, DecDEC proxy, and M11 have lowest residuals under the sublinear square-root fit among the tested alternatives. Their mean KL values are 0.150, 0.133, and 0.131 respectively. Their fitted AR(1) residual decays in

$$D_t - \hat{D}_t = \rho(D_{t-1} - \hat{D}_{t-1}) + \eta_t$$

are 0.51, 0.45, and 0.44. Appendix D reports the R^2 table.

The narrowed cross-model diagnostic gives the same qualitative answer on the two smaller non-Granite models. DeepSeek-R1-Distill and Falcon-H1 both select sublinear square-root fits for static top-1%, DecDEC proxy, and M11. Their activation spectra are broadband rather than periodic: median normalized spectral entropy is 0.850 for DeepSeek and 0.891 for Falcon, with 100-token autocorrelation in both packets. This weakens compound error

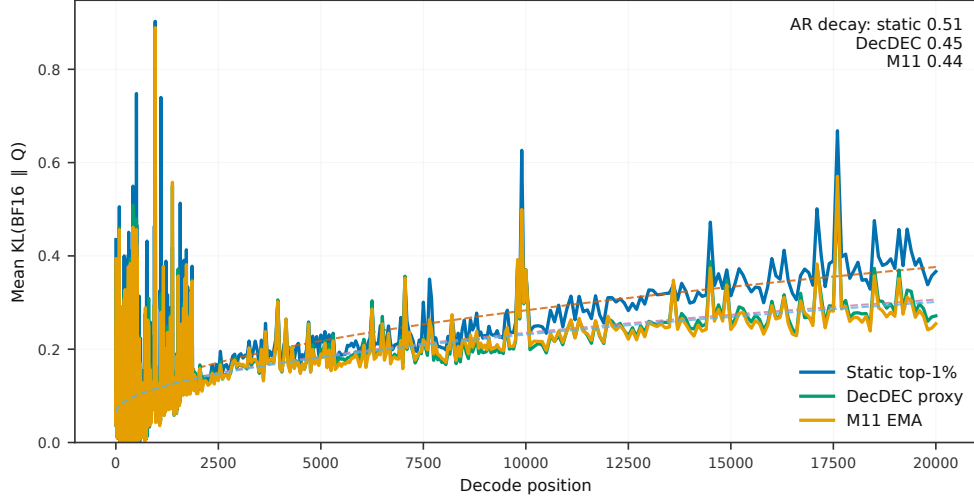


Figure 4: Per-position KL trajectories on Granite-Small. Among the tested functional forms, all three regimes have lowest residuals under sublinear square-root fits and moderate AR decay, rather than superlinear growth.

as the primary explanation in the measured regime and shifts attention back to stale or misbudgeted protection.

6 Discussion

The results point to three mechanisms. A simple error-allocation model makes the link from set-leaving to recovery explicit. Let $\epsilon_p \ll \epsilon_q$ be the average error from protected and unprotected important channels. If a static policy protects $S(t_0)$ while the current important set is $S(t)$ and $|S(t)| = K$, then

$$\mathbb{E}[\text{err}(t)] \approx \epsilon_p |S(t_0) \cap S(t)| + \epsilon_q |S(t) \setminus S(t_0)| = K \{\epsilon_p (1 - L(t_0, t)) + \epsilon_q L(t_0, t)\}.$$

The term multiplied by $L(t_0, t)$ is the cost of allocating protected capacity to channels that have left the current top- K set.

First, boundary discontinuities are actively harmful: both hard protected-set and hard scale-bin switching lose to random controls. Salfati’s KV-cache analysis provides a compatible formal analogy: under the softmax Fisher metric, dropping a direction is quadratically worse than quantizing it at the same bit cost (Salfati, 2026). Second, smoothing removes the worst boundary harm but does not recover quality at top-1% budget. M11 and M18 contain signal, but the union of important channels across a long trace exceeds the fixed top-1% capacity. Third, budget is binding. Expanding K from 1% to 10% gives EMA enough capacity to track a larger portion of the moving set on Nemotron, but budget alone is not a universal recipe: Nemotron top-10 succeeds, Granite top-5 recovers 0.449 with wide uncertainty, and static top-10 is already strong on Nemotron.

Trace heterogeneity remains important. In most Granite static-top-1% channel-set experiments, roughly one third of the 12-trace slice has no recoverable static gap under the ratio metric; in the larger static-union experiment the no-gap fraction is 0.375. Early static-union gates retain no-gap traces as zero-recovery rows, while later method packets report recovery ratios on positive-gap traces and track the no-gap fraction separately. This changes the estimand across early gates and later positive-gap packets, so we do not compare their medians as if they were identical population quantities. It also explains the wide CIs on several methods. After BCa/Holm correction over 33 method-family tests, statistically positive channel-set evidence is limited to Nemotron budgeted EMA top-5/top-10; paired method-vs-control margins remain descriptive unless separately tested. The paper empha-

sizes controlled mechanism comparisons and cross-model directionality rather than a single universal method headline.

The saturated branches now read as design constraints. Hard position switching loses to random controls, so discontinuous set changes should be avoided. Top-1% EMA has signal but low recovery, so the top-1% budget is under-sized for long traces. M-PRED fails badly on Granite and does not rescue DeepSeek or Falcon, consistent with broadband drift entropy of 0.85–0.89 defeating simple prediction. ParoQuant+M11b is sub-additive, so rotation and channel protection are not automatically compositional. Falcon resists EMA, so architecture-local validation is required before claiming a channel-set remedy. ParoQuant shows that rotations can recover much of the Granite W4A16 gap; M11b shows that protected budget matters on Nemotron but does not transfer unchanged to DeepSeek or Falcon. The practical outcome is regime-aware selection from measurable model properties, plus future work on methods that couple rotation and channel selection more tightly.

The natural coupling is in the rotated residual basis. Original-basis M11b ranks channels by a moving estimate of $x_i(t)^2$ and fails when channel identity drifts. ParoQuant rotates the basis and quantizes a fixed transformed weight. If W_{pq} is the dequantized rotated weight and $\Delta W = W_{fp} - W_{pq}$, a residual-correction policy would select columns by a score such as

$$s_i(t) = \text{EMA}(x_i(t)^2) \|\Delta W_{:,i}\|^2$$

and apply the continuous correction $x_P(\Delta W)_P$ for a small working set P . This keeps the useful part of M11b’s tracker but moves the target from original-basis channel identity to post-rotation residual benefit. The current top-8x32 residual proxy does not pass the Granite tail gate even though it selects valid post-rotation residual columns. That negative result is a mechanism finding: residual norm alone is not a reliable surrogate for output-distribution benefit. Any future residual method must first show a large KL-lookahead headroom, select a column set materially different from the killed proxy, and confirm the gain on held-out traces before it can replace the regime protocol.

The same residual screen also gives a useful systems bound. For m corrected modules, e active experts, output width r , corrected input columns k , and element size b bytes, the additional work is

$$\text{MACs/token} \approx \text{merk}, \quad \text{HBM/token} \approx \text{merk}b, \quad E_{\text{token}} \approx B_{\text{HBM}}\epsilon_{\text{HBM}} + \text{MACs}\epsilon_{\text{MAC}}.$$

The hot working set is the sum of stored residual sidecars, $\sum_j r_j k_j b$. The tested top-8x32 Granite sidecar stores 54.0 MiB of FP16 residual columns. With 10 active experts, 1536 output rows, 32 corrected input columns, and eight corrected MoE input modules, a naive correction would add roughly 3.9M multiply-accumulates per generated token and stream about 7.5 MiB/token of active residual columns if the sidecar is not cache-resident. This aggregate estimate is the per-projection envelope multiplied by the corrected modules and active experts. It assumes no useful cross-token cache residency for the active sidecar; batching or locality could reduce HBM traffic, while gather/scatter and launch overheads could increase latency. Using a simple device-agnostic 10–20 pJ/byte HBM and 1–5 pJ/MAC analytical cost model, that is on the order of 83–177M pJ/token before kernel overhead. Because the quality gate fails, we do not implement the kernel or report profiler measurements; the analytical estimate records what a successful residual method would need to justify. Appendix C reports the per-projection envelope used to generate the artifact cost table.

7 Limitations

We scope the study to W4A16 PTQ on small reasoning models and measured activation-channel surfaces. We do not claim results for FP8, MXFP4/NVFP4, W4A4, or quantization-aware training; the format-axis probe found available FP8/W4A16 paths but no benchmark-metrics adapter, and no importable NVFP4 runtime kernel. The hybrid-model Quamba2 comparison is mainly at the accessible block-output surface plus one Granite internal-surface diagnostic. That internal packet covers SSM input, B/C generation, Mamba out-projection input, attention output-projection input, and post-block output, but only on two Granite

traces; Falcon branch-local internals remain future work. Granite-Small has the densest FFT and KL trace, while DeepSeek and Falcon have narrowed KL/FFT replications; Nemotron dense diagnostics are deferred because throughput would have dominated the Stage 1 budget. Qwen3 scale-up is also deferred because the planned W4A16 checkpoint snapshot was not present locally. The M11b recovery result is model-dependent: Nemotron top-10 succeeds, Granite’s strongest signal is top-5 with wide uncertainty, and DeepSeek/Falcon top-10 vetting is ambiguous. The strongest current remedy is a ParoQuant-style rotation baseline, which we do not claim as our method; our additional design evidence is limited to the regime protocol and the negative DriftRot screens reported here. The practical recommendation is to measure drift, tail risk, recoverable gap, and rotation signal before choosing channel-set protection, rotation, tail-control, or rejection of channel-set protection for a new model.

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Ethics Statement

This work analyzes benchmark prompts and model activations. It does not use private user data or make deployment-safety claims. The main ethical risk is overclaiming a scoped mechanism study as a universal quantization recipe; the paper separates measurement, mechanism, and partial-remedy claims.

Reproducibility Statement

The release package contains the public FAST_VERIFY interface, including release/README.md, release/docs/reproducing_results.md, and release/src/scripts/reproduce_all.sh. FAST_VERIFY replays frozen paper-claim values and validates the documented tolerance mapping. Full-fidelity GPU reruns are not implemented inside release/; they remain in the archived experimental packet runners and are disclosed as a release limitation. Stage 1 claim-bearing packets now include the MATH-500 drift replication, DeepSeek/Falcon KL/FFT diagnostic, Granite composition test, BCa/Holm correction, ParoQuant baseline vetting on Nemotron/DeepSeek/Falcon, and DriftRot tight-clip/residual/M-SURFACE screens. The format-axis and Qwen3 packets are retained only as infrastructure provenance.

Model	HuggingFace identifier	Snapshot commit
Granite-Small	ibm-granite/granite-4.0-h-small	b8c0982b
Nemotron-3	nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B-BF16	cbd3fa9f
DeepSeek-R1-Distill	deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B	ad9f0ae0
Falcon-H1	tiiuae/Falcon-H1-0.5B-Instruct	8f2587ca

Table 2: Model checkpoints used by the experiments. Full provenance and file-level hashes are recorded in the archived result manifests; the release package records claim-to-script tolerance mappings.

Bootstrap seeds, deterministic trace indices, method budgets, and tolerance windows are specified in the release configs. The E15 packet records the 33-test Holm family explicitly in e15_bca_holm.json: each row lists packet, regime, raw one-sided median-recovery p value, Holm rank, adjusted p value, threshold, and reject flag. GPU kernels can still exhibit minor nondeterminism across driver versions; the verification compares outputs to bootstrap CI tolerances rather than bit-exact scores. The original Qwen3.6 exploratory hook surface was not used for paper claims because activation access was incomplete.

LLM Use Disclosure

Following the COLM 2026 LLM-use policy adopted from ICLR 2026, we disclose substantive use of large language models in preparing this work. Anthropic Claude and the Codex autonomous coding agent were used to draft and revise paper prose, including the abstract, related work, methods, and results sections; to generate the matplotlib code used for Figures 1–4; to conduct iterative simulated peer review and audit passes; to summarize local experimental artifacts; and to propose, prioritize, and critique candidate experimental methods and follow-up branches, including method families reported in this paper. Codex was also used as a programming assistant for experiment runners, checkers, release-package code, and verification scripts. The human author retained responsibility for final experiment authorization, numerical interpretation, scope decisions, and submission framing. All paper references and attributed prior-work claims were checked against primary sources in the hallucination audit summarized in Appendix F; a later user-flagged scoop check added ResQ, Salfati’s KV-cache analysis, and CMPQ after arXiv verification. All reported experimental numbers were traced to committed score caches or result artifacts rather than accepted from LLM output.

A Detailed Experimental Provenance

Paper description	Internal study label	Provenance location
Granite-Tiny pilot drift measurement	Phase 0	experimental/outlier_migrate/phase0/
Granite-Small drift replication	Phase 1	experimental/outlier_migrate/phase1/
Nemotron-3 drift replication	Phase 2	experimental/outlier_migrate/phase2/
Static migration-aware union, Granite-Tiny	Phase 3	experimental/outlier_migrate/phase3/
Static migration-aware union, Granite-Small	Phase 4	experimental/outlier_migrate/phase4/
DeepSeek Transformer control	Phase 5 prime	experimental/outlier_migrate/phase5_prime/
Falcon-H1 parallel-hybrid control	Phase 6	experimental/outlier_migrate/phase6/
Channel-set methods and baselines	Phase 7	experimental/outlier_migrate/phase7/
MATH-500 drift replication	Method study	experimental/outlier_migrate/phase7/
Cross-model KL/FFT diagnostic	Stage 1 E2	experimental/outlier_migrate/phase7/results/om_stage1_e2_mathonly_20260526T1202Z/
ParoQuant-M11b composition	Stage 1 E1	experimental/outlier_migrate/phase7/results/om_stage1_e1_deepseek_falcon_20260526T2335Z/
Format-axis probe	Stage 1 E3	experimental/outlier_migrate/phase7/results/om_stage1_e3_granite_20260526T0344Z/
Qwen3 scale-up probe	Stage 1 E4	experimental/outlier_migrate/phase7/results/om_stage1_e4_granite_probe_20260527T093400Z/
BCs/Holm correction	Stage 1 E5	experimental/outlier_migrate/phase7/results/om_stage1_e5_holm_20260527T093400Z/
ParoQuant-on-Nemotron vetting	VI extension	experimental/outlier_migrate/phase7/results/om_v1_paroquant_nemotron_20260528T0318Z/
ParoQuant DeepSeek/Falcon vetting	VI extension	experimental/outlier_migrate/phase7/results/om_v1_paroquant_deepseek_20260528T162800Z/
DriftRot clip	DriftRot clip	experimental/outlier_migrate/phase7/results/om_driftrot_granite_clip_tight_finalpoor_20260528T2251Z/
DriftRot tight-clip Granite confirmation	DriftRot clip	experimental/outlier_migrate/phase7/results/om_driftrot_granite_clip_tight_20260528T2314Z/
DriftRot tight-clip DeepSeek/Falcon transfer	DriftRot clip	experimental/outlier_migrate/phase7/results/om_driftrot_deepseek_clip_tight_20260528T2314Z/
DriftRot residual correction diagnostic	DriftRot residual	experimental/outlier_migrate/phase7/results/om_driftrot_residual_granite_tail_diag_fixed_20260528T2116Z/
Granite internal-surface diagnostic	MSURFACE	experimental/outlier_migrate/phase7/results/om_phase9_surface_granite_internals_20260529T000558Z/
M11b DeepSeek/Falcon vetting	V2	experimental/outlier_migrate/phase7/results/om_v2_m11b_deepseek_20260527T1310Z/
Paper release interface	Release package	release/docs/reproducing_results.md

Table 3: Translation from paper-facing experiment names to internal study labels. Exact packet directories, command lines, prompt hashes, and model file hashes are maintained in the release documentation and result manifests.

The valid Nemotron budget-tuning result uses the corrected static top-1% baseline rerun and reuses checked score caches for the budget arms. The earlier invalid baseline is excluded from all recovery claims. The release mapping table records both the valid rerun and the excluded failed attempt so that reviewers can audit the correction.

Stage 1 packets E1, E2, E3, and E15 are claim-bearing. E4 and E5 are infrastructure-disclosure packets only: E4 did not produce benchmark accuracy/throughput/VRAM rows, and E5 was skipped because the planned local Qwen3 W4A16 snapshot was unavailable.

B Per-Method Detailed Results

C Systems Cost Envelope

D KL Fit Details

E Extended Layer-Stratified Results

F Hallucination and Source Audit

Before finalizing this draft, we audited every bibliography entry against its primary source URL and checked the attributed prior-work claims in the main text. The audit removed uncited bibliography entries, kept ChanMix author names only after checking the public OpenReview page, and tightened wording for DecDEC, Quamba2, ParoQuant, Rotated Runtime Smooth, ResQ, Salfati, Activation Sensitivity, and SLQ so that

Experiment	Included traces	No-gap fraction	Median recovery	95% CI
Static union, Granite-Tiny	24	0.417	0.000	[0.000, 0.711]
Static union, Granite-Small	24	0.375	0.000	[0.000, 0.0695]
Position-conditioned channels	8	0.333	-0.867	[-4.25, 0.357]
Position-binned scales	12	0.000	0.234	[-3.47, 0.490]
EMA-smoothed top-1%	8	0.333	0.048	[-8.94, 0.448]
Activation+key coupling	8	0.333	-0.344	[-14.1, 0.731]
DecDEC proxy	8	0.333	-0.070	[-8.50, 0.391]
Budget EMA top-5, Granite	8	0.333	0.449	[-2.28, 1.00]
Static top-10, Nemotron	10	0.167	0.594	[0.246, 0.814]
Budget EMA top-5, Nemotron	10	0.167	0.457	[0.313, 0.760]
Budget EMA top-10, Nemotron	10	0.167	0.815	[0.158, 0.906]
Budget EMA top-10, DeepSeek	11	0.083	0.335	[-0.409, 0.494]
Budget EMA top-10, Falcon	12	0.000	0.044	[-0.144, 0.203]
Stable-core protection	8	0.333	0.178	[-1.78, 0.986]
ParoQuant rotation	8	0.333	0.754	[0.477, 1.00]
ParoQuant rotation, Nemotron	10	0.167	1.05	[1.01, 1.29]
ParoQuant rotation, DeepSeek	11	0.083	0.756	[-0.246, 0.855]
ParoQuant rotation, Falcon	12	0.000	0.381	[0.0645, 0.547]
ParoQuant+M11b top-10	8	0.333	0.565	[-63.6, 0.932]
DriftRot tight clip, Granite pos-gap	8	0.000	0.922	[0.781, 1.65]
DriftRot tight clip, DeepSeek	11	0.083	0.518	[0.265, 0.797]
DriftRot tight clip, Falcon	12	0.000	0.390	[0.0019, 0.637]

Table 4: Detailed rounded outcomes for all method families discussed in the main text. Values are recovery ratios relative to each experiment’s BF16-vs-static baseline.

Screen	Status	One-line reason
Static top-1% / static union	KILL	strict leaving reaches 53–67%; static-union medians are zero on Granite packets
Hard position switching	KILL	position-conditioned sets lose to the random-bin control by 0.668 median recovery
Position-binned scales	KILL	hard scale bins lose to the random scale-bin control by 0.761 median recovery
M11 EMA top-1%	KILL	EMA has signal over random walk, but absolute recovery is only 0.048
M11b budget EMA	AMBIG / scoped pass	Nemotron top-10 passes; Granite is wide-CI and DeepSeek/Falcon are ambiguous
M-PRED predictor	KILL	predictive timing adds negative tails and does not rescue DeepSeek or Falcon
WJAC sensitivity proxy	KILL	weight-norm proxy is nearly collinear with EMA scores and changes little
K-RES residual proxy	KILL	valid top-8x32 rotated-basis residual correction worsens the Granite tail gate
DriftRot tight clip	AMBIG	fixes Granite positive-gap tail but does not transfer cleanly to DeepSeek or Falcon
M-SURFACE	KILL	complete Granite internals find no surface below the promotion threshold
ParoQuant-style rotation	PASS baseline	strongest prior-work remedy class across the measured model set; not our method
M-BRANCH / BranchRot	FUTURE WORK	Falcon branch-local cache was not captured in this pass

Table 5: Final method-screen status table. Positive-method search is stopped for this submission cycle; failed branches are reported as mechanism evidence rather than hidden tuning attempts.

the prose matches the source claims. No cited bibliography entry is missing, orphaned, or unresolved after these corrections. We separately re-derived the headline experimental numbers from local score caches and result manifests; the audit identified only word-ing/rounding fixes, not a change to any reported result. The full audit table is maintained in docs/hallucination_audit_report.md and references_audit.md.

Method/question	Static vs decode-time	Rotation vs channel-set	Long-decode drift?	Positive-method claim?	Safe distinction	Unsafe claim to avoid
ParaQuant	static calibration	rotation	no	yes, prior work	our packets use it as strongest rotation baseline	claiming ParaQuant as our method
QuatRot	static calibration	rotation	no	yes, prior work	shows rotations can remove activation outlier difficulty	claiming long-reasoning drift measurement
SpinQuant	learned/static rotation	rotation	no	yes, prior work	learned rotations are adjacent basic-conditioning methods	claiming we invent rotation quantization
MambaQuant	static PTQ	rotation/channel smoothing	no	yes, prior work	Mamba-family quantization baseline for static calibration	saying it studies long decode protected-set drift
Quant2	static/online SSM structure	channel/state grouping	not at our horizon	yes, prior work	its internal persistence does not imply block-output top-k stability	saying Quant2 is wrong at all internal tensors
DecDEC	decode-time	channel-set selection	short horizon	yes, prior work	qualitative precursor for dynamic salient channels	claiming our reactive selection is DecDEC itself
This paper	decode-time measurement	channel-set tests plus rotation baseline	yes	no universal quantizer	long-decode drift, controlled failure taxonomy, regime protocol	claiming a new globally positive quantizer

Table 6: Novelty and scope boundary against the closest rotation, Mamba quantization, dynamic-channel, and SSM quantization work.

Granite surface	Mean strict leaving	Median strict leaving	Delta vs post-block mean
Post-block output	0.546	0.537	0.000
SSM input after convolution	0.484	0.512	-0.061
SSM C generation after convolution	0.486	0.500	-0.060
SSM B/C generation concat	0.602	0.667	+0.056
SSM B generation after convolution	0.618	0.500	+0.072
Mamba out-projection input	0.887	0.902	+0.341
Attention output-projection input	0.799	0.793	+0.250

Table 7: Two-trace M-SURFACE diagnostic with SSM internals. SSM input and C generation drift slightly less than post-block, but neither is below 0.30–0.40 or at least 0.15 below post-block, so the complete Granite surface screen does not promote surface-local protection.

Model	AIME set-leaving	MATH-500 set-leaving	Delta
Granite-Small	0.566	0.561	-0.005
DeepSeek-R1-Distill	0.671	0.651	-0.020
Falcon-H1	0.674	0.675	0.002

Table 8: Stage 1 E2 cross-prompt drift replication. MATH-500 rates are computed on 30 prompts per model and all fall within the fixed ± 0.15 tolerance from the AIME-2025 reference.

Representative width	D_{out}	K	Fused HBM/token	MACs/token	Analytical energy/token
DeepSeek width	1536	8	24.0 KiB	12.3K	0.258–0.676M pJ
Falcon width	2048	8	32.0 KiB	16.4K	0.344–0.902M pJ
Nemotron width	2688	8	42.0 KiB	21.5K	0.452–1.18M pJ
Granite width	4096	8	64.0 KiB	32.8K	0.688–1.80M pJ
Granite top-8x32 active aggregate	1536	8×32	7.5 MiB	3.9M	83–177M pJ

Table 9: Analytical envelope for a hypothetical residual-column correction with fp16 residual weights, fp16 activation values, and uint16 indices. The first four rows are per projection at $K = 8$; the aggregate row multiplies the Granite MoE packet by eight corrected modules, 10 active experts, and 32 corrected columns. The lower energy bound uses fused output update with 10 pJ/byte and 1 pJ/MAC; the upper bound includes output read/write with 20 pJ/byte and 5 pJ/MAC. These are estimates, not profiler measurements.

Regime	$R^2_{\sqrt{t}}$	R^2_t	R^2_{ρ}	AR(1) ρ
Static top-1%	0.469	0.454	0.380	0.515
DecDEC proxy	0.409	0.379	0.298	0.446
M11 EMA	0.399	0.368	0.284	0.439

Table 10: Granite-Small per-position KL functional-form fits. The square-root fit has the highest R^2 for each tested regime, while the residual AR(1) coefficient remains moderate rather than near one.

Model	Layer type	Layers	Strict set-leaving	Within-set shuffling
Granite-Tiny	Attention	4	0.618	0.186
Granite-Tiny	SSM/Mamba	36	0.636	0.174
Granite-Small	Attention	4	0.557	0.282
Granite-Small	SSM/Mamba	36	0.567	0.270
Nemotron-3	Attention	6	0.563	0.255
Nemotron-3	MoE	23	0.527	0.278
Nemotron-3	SSM/Mamba	23	0.533	0.264

Table 11: Layer-type decomposition. Strict set-leaving is the component most directly relevant to protected-set membership; within-set shuffling motivates scale-refresh mechanisms but is not itself membership loss.